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SURVEY

A Survey of Autonomous Robotic Ultrasound Scanning Systems

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ABSTRACT This review investigates recent advancements in autonomous, semi-autonomous, and teleoperated robotic ultrasound systems. Traditional ultrasound imaging depends on manual probe manipulation, which introduces operator variability, physical strain, and limitations in accessibility. To address these challenges, this review investigates recent advancements in autonomous, semi-autonomous, and teleoperated robotic ultrasound systems by analyzing over 60 publications, including key developments from 2022 to 2025. Our survey reveals a growing adoption of cobot-based solutions equipped with 6-DOF force/torque sensors and RGB-D vision systems for precise probe positioning. Notably, several systems now integrate reinforcement learning, image-guided visual servoing, and real-time feedback loops to enable intelligent trajectory planning and adaptive force control. However, we identify critical gaps in the literature: surface-parallel force and torque components are often ignored in control models, limiting the accuracy of probe orientation and tissue coupling. Furthermore, real-time ultrasound image feedback is rarely used for path optimization, despite its importance in enhancing image quality and diagnostic reliability. This review emphasizes the need for future systems to integrate multi-modal sensing, adaptive control, and real-time image quality assessment to achieve robust, generalizable robotic ultrasound workflows.

INDEX TERMS Computer-aided systems, echocardiography, deep learning, medical robotics, neural networks, robotic system and software, robotic ultrasound, ultrasound imaging.

I. INTRODUCTION

Ultrasound imaging has been a cornerstone of medical diagnostics since its introduction in the 1940s and is now one of the most widely used imaging modalities worldwide [1], [2], [3], [4]. Its advantages, including non-invasiveness, cost-effectiveness, and real-time imaging capabilities, make it indispensable in various medical fields such as obstetrics and gynecology [5], [6], urology [7], [8], neurology [9], [10], [11], [12], and cardiology [13], [14], [15], [16], [17].

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To address growing demands for scalable, consistent, and operator-independent imaging, autonomous and robotic ultrasound systems have emerged as a promising solution. These systems aim to automate probe manipulation, integrate real-time feedback, and reduce reliance on skilled human operators. This survey focuses specifically on the development, capabilities, and clinical relevance of such robotic ultrasound platforms.

Recent advancements in robotic ultrasound imaging have spanned a wide range of applications, from fully autonomous systems to co-robotic and tele-operated platforms. As shown in Table3, the literature between 2022 and 2025 highlights emerging trends in automation, haptic feedback, and

AI-driven optimization strategies. Notably, Su et al. [35] introduced a fully autonomous robotic ultrasound system capable of thyroid scanning with reinforcement learning and human-skeleton recognition, while Lin et al. [36] proposed Auto-RUSS, a robotic system optimized for reproducibility in image acquisition. Similarly, Najdovski et al. [38] explored haptically enabled tele-ultrasound for remote diagnostics, and Antolin et al. [40] validated a 5G-based system through clinical trials. Although these systems demonstrate promising capabilities, many studies remain constrained to organ-specific evaluations, phantom-based simulations, or small-scale cohorts. Moreover, reviews like Du et al. [39] and Bi et al. [31] emphasize the lack of integration between adaptive force control and real-time image feedback mechanisms. This growing body of work signals the need for comprehensive solutions that unify sensing, learning, and clinical validation—paving the way for reliable and generalized autonomous robotic ultrasound systems.

Traditional ultrasound scanning relies on trained sonographers to manually adjust the probe's position, angle, and pressure to capture accurate images. However, this dependence has led to significant challenges, including a growing shortage of skilled sonographers, especially in rural and remote areas. Additionally, the repetitive physical demands of ultrasound scanning contribute to occupational musculoskeletal disorders among sonographers, particularly in the neck, shoulder, and wrist regions [19], [20], [21]. Furthermore, the close proximity required between the sonographer and the patient poses a heightened risk of infection, especially during pandemics such as COVID-19 [22], [23]. These challenges highlight the urgent need for innovative solutions such as remote or automated ultrasound imaging technologies.

To address these limitations, various teleoperated ultrasound systems have been explored, particularly during the COVID-19 pandemic, where remote ultrasound assessments of vascular, cardiac, and pulmonary conditions were facilitated by advanced communication networks like 5G [24], [25], [26]. Furthermore, robotic-assisted ultrasound has been introduced for specialized applications, such as robotic transcranial Doppler systems and automated breast ultrasound imaging. Automated breast ultrasound systems, for instance, have significantly improved early breast cancer detection by reducing operator dependency and enhancing diagnostic efficiency [27], [28].

Several recent surveys have substantially advanced the field's understanding of robotic ultrasound. Jiang et al. [29] present a systems-level overview of robotic ultrasound platforms, while Salcudean et al. [30] discuss the broader field of robot-assisted medical imaging. Bi et al. [31] and Li et al. [32] focus on learning-driven automation and adaptive path planning, and von Haxthausen et al. [33] provide insight into design architectures and trends. Our review builds upon these works by concentrating specifically on the enabling control and perception technologies—such as real-time image-guided control, lateral force/torque feedback, and hybrid sensing strategies—that are being deployed in experimental and preclinical systems developed between 2022 and 2025.

Robotic ultrasound, which integrates robotic assistance with ultrasound imaging, is gaining traction as a viable alternative for overcoming the limitations of manual scanning. Robotic systems offer high precision, dexterity, and repeatability, making them well-suited for automating ultrasound procedures [34]. While progress has been made, research is still ongoing to enhance the adaptability, accuracy, and accessibility of robotic ultrasound technology across diverse clinical applications.

This paper aims to provide a comprehensive overview of recent advancements in robotic ultrasound, analyzing its potential to revolutionize diagnostic imaging by addressing workforce shortages, improving accessibility, and enhancing diagnostic accuracy. Several studies have reviewed autonomous robotic ultrasound scanning systems; however, most existing reviews focus primarily on broad overviews of robotic ultrasound technologies without an in-depth analysis of force control mechanisms, real-time feedback integration, and image-driven optimization. The uniqueness of this review lies in the following key aspects:

- A survey of force control strategies in robotic ultrasound systems across different levels of autonomy (teleoperated, semi-autonomous, and fully autonomous), highlighting how these methods maintain appropriate contact force during scanning.
- A discussion of the role of lateral (surface-parallel) force components and probe torque in maintaining stable probe contact, including strategies to control these factors for consistent ultrasound imaging.
- An overview of ultrasound image-based guidance and optimization techniques, describing how real-time imaging feedback is used to refine probe placement with the aim of improving diagnostic precision.
- This survey goes beyond general overviews by providing a focused analysis of key technical components critical to autonomous robotic ultrasound systems.

A. CLARIFICATION OF TERMINOLOGY: TELEOPERATED VERSUS ROBOT-ASSISTED ULTRASOUND SYSTEMS

We acknowledge the need for greater clarity in distinguishing between *teleoperated* and *robot-assisted* ultrasound systems. In this review, we define **robot-assisted ultrasound systems** as any system where a robotic manipulator is used to support, position, or manipulate the ultrasound probe. These systems range from manually guided robots to those employing full or partial automation. Within this category, **teleoperated systems** represent a specific subset wherein the robotic manipulator is operated remotely by a human user, often via haptic interfaces or real-time communication links [40], [43].

The key distinction lies in the locus of control: teleoperated systems rely entirely on human operators for decision-making and probe manipulation, albeit at a distance, while other robot-assisted systems may incorporate autonomous features such as trajectory planning, force control, or image-guided optimization [38], [39]. For instance, the TER system [59] and more recent 5G-based platforms [44] are teleoperated systems, whereas platforms like Auto-RUSS [36] or ARUS [48] involve substantial autonomous functionalities even though they fall under the broader umbrella of robot-assisted ultrasound.

To reduce ambiguity, throughout this paper we have used the term *robot-assisted ultrasound* as an umbrella category and specify *teleoperated*, *semi-autonomous*, or *autonomous* when the control paradigm is relevant to the discussion.

II. AUTONOMY OF ULTRASOUND ROBOTS

Robotic autonomy is defined by different levels, depending upon the improvised degree of operational ease, user independence, and reduction in operators' burden. The level of autonomy of robot systems ranges from 0 to 3 depending on their level of independence from the operator (Refer to Figure 1).

A. AUTONOMICAL LEVEL 0

This level is known as the manual probe manipulation and is referred to as 0 or no autonomy. Probe movement is directed by the user, and the ultrasound probe, which is held by the robot, automatically calculates the actions to complete that motion. There are two types of systems at this level: the first is tele-echography [58], [59], and the second one is teach/replay [60], [61]. In the first type, a sonographer residing in a remote area can operate the robot in real-time by using some device such as a 3D mouse or a joystick for medical examinations [62], [63]. In the latter type, the probe is attached to a robot-arm and all movement of the arm is tracked. After learning these trajectories, the robotics system present at the patient's end is able to undergo the examination procedure by following the directions learned from the ultrasound probe [64], [65]. When we compare these two systems, we find that "teach/replay" gives more advantages as it omits the complicated procedure of controlling the robot remotely, which requires the transfer of ultrasound images in real-time, hence decreasing the amount of required bandwidth and communication necessity during the remote assessment [64]. This only works if the patient does not move. The learning procedure may be necessary for each individual patient. However, this cannot be avoided in clinical practice. As the movement of the probe is entirely controlled by the user in these systems, the muscular strain of the sonographer is considerably reduced when moving with telemetric movement, resulting in the quality of images being solely dependent on the person operating the probe. This remote control procedure contains complications that might result in overloading information on the other end where the task execution has its bounds [32], [65].

B. AUTONOMICAL LEVEL 1

This level is referred to as "human-robot cooperation" or assistance to the robot [66]. This arrangement provides the sonographer and the robot with equal control over the movement of the probe. In teleoperated ultrasonography, a joint control over a two-dimensional ultrasound probe employing three degrees of freedom was first performed by Abolmaesumi et al. [67], [68] and Salcudean et al. [69], [70], [71] by use of visual servoing dependent on images. Visual servoing enables the robot to trace the required features from the image by itself, resulting in the reduction of the unnecessary movements of the medical subjects at the time of teleoperation, therefore minimizing the operator's responsibilities to some extent, making it less user-dependent. Sen et al. [72] came to aid the operators who lack expertise in acquiring ultrasound images by inventing a cooperative control strategy with the help of a virtual fixture for radiotherapy procedures. A dual force sensing system was developed by Fang et al. [73] to go along with the operator's movement at the time of stable contact force undergoing the admittance control to take the edge off the sonographer. For a better assessment of this force assistance on the stability of ultrasound images, a study was conducted based on users by Finocchi et al. [74]. Human-robot systems enable novice operators to assist ultrasound scanning, unlike fully teleoperated systems. Continuous communication lets on-site clinicians ensure patient safety and adjust the procedure as needed [32], [73]. These advantages make them more likely to be implemented in clinical practice.

C. AUTONOMICAL LEVEL 1.5

These systems are named "semiautomated systems," which means "a system that can fully automate the probe movement but necessitates initial position to be fed manually." For instance, a system that stabilizes autonomously by using visual servoing for an already selected point on the image [32], [75], [76], [77]. Due to the involvement of manual operations, these are named semi-automated systems.

D. AUTONOMICAL LEVEL 2

As these systems undergo the path defined manually for autonomous robotic ultrasound imaging, therefore they are named "task-defined" systems. Operators can dictate the scanning trajectory at the time of planning. Tissue deformation and the expected patient movement can put some limitations on the flexibility of the scanning trajectory. To counter these limitations, online manipulations with the help of attained sensor data and automatic adjustments to the predefined path are utilized.

E. AUTONOMICAL LEVEL 3

In the absence of any human-defined instruction, these systems can perform the planning and implementation of the ultrasound image acquisition; however, all of this will be done under the operator's supervision, and

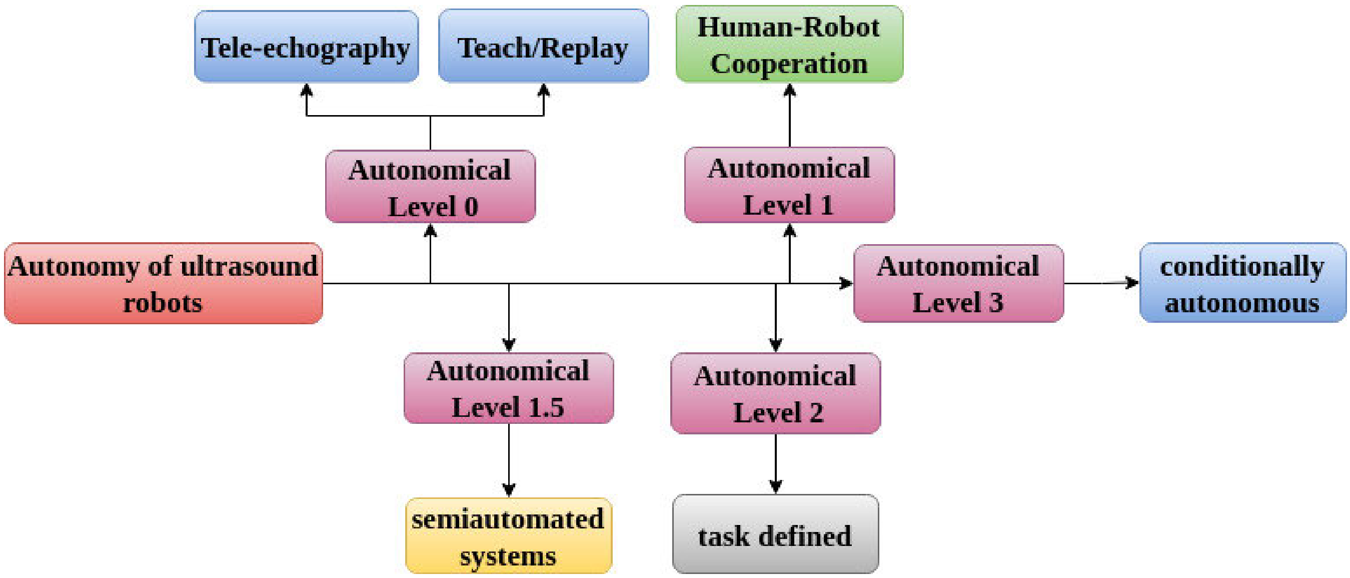

FIGURE 1. Categorization of ultrasound robot systems based on their autonomy levels.

TABLE 1. Selection criteria (For the identification of articles found in the literature to be included in this survey).

No.	Inclusion	Exclusion
1	Ultrasound image acquisition by autonomous robots for medical use, i.e., 1. Biometric measurement 2. Diagnosis 3. Interventional support	Autonomous image acquisition for non-medical use for modalities, i.e., 1. Computed tomography 2. X-rays 3. Magnetic Resonance Imaging
2	Original/translated English Language articles	Neither original not translated in English articles
4	proper description of systems	Only ultrasound image analysis
5	autonomous ultrasound images	simulated ultrasound images
6	Presence of physical robot	No robotic system
7	Detailed methodology and system explanation	No Detailed methodology and system explanation
8	Fully automatic system	Partially automatic
9	No human interventions	Full partial human intervention
10	Acquisition result shown	No acquisition result shown

hence it is called "conditionally autonomous." These systems can directly entertain the sensor data and change accordingly, thereby providing maximum assistance to the sonographer.

There are many review articles that focus mainly on levels 0, 1, and 1.5, whereas this article focuses on levels 2 and level 3 [34].

F. SEARCH METHODOLOGY

To ensure a comprehensive and up-to-date review, we conducted a systematic literature search of robotic ultrasound systems using four primary academic databases: IEEE Xplore, PubMed, ScienceDirect, and SpringerLink. The search was conducted in two phases: an initial search in July 2024 and an update in February 2025. We included articles published from January 2022 to March 2025.

TABLE 2. Categorization of recent robotic ultrasound studies (2022–2025).

Study	Force Control	Image Optimization	Teleoperation / Haptics
Su et al. (2024) [35]	✓	✓	
Lin et al. (2025) [36]	✓		
Najdovski et al. (2024) [38]	✓		✓
Raina et al. (2023) [46]		✓	
Zhang et al. (2024) [47]		✓	
Planes et al. (2025) [40]			✓

We used a combination of keywords and Boolean logic to capture relevant studies: (robot* OR autonomous OR teleoperation) AND (ultrasound OR sonography). The search was applied to multiple fields including **titles, abstracts, and keywords**. In several databases (e.g., ScienceDirect and Springer), we also searched the **full text** to avoid excluding papers that discuss robotic ultrasound systems within broader automation contexts.

This search strategy yielded an initial set of 263 articles. After removing duplicates and screening based on relevance and publication type (excluding non-peer-reviewed content, short abstracts, and non-English papers), we retained 86 articles for full-text review. Based on their direct relevance to robotic ultrasound applications—including probe control, force regulation, trajectory planning, feedback integration, and diagnostic performance—**43 core studies** were selected for detailed synthesis. In response to reviewer feedback, we also manually included recent relevant articles not

TABLE 3. Comparison of recent reviews on robotic ultrasound systems (2022–2025).

Reference	Focus Areas	Limitations
Su et. al. (2024) [35]	**Fully autonomous thyroid ultrasound scanning:** Develops a fully autonomous robotic system for thyroid exams, using human skeleton point recognition, reinforcement learning, and force feedback to locate the thyroid and adjust probe orientation.	**Organ-specific and initial study:**Demonstrated only for thyroid scanning on a limited participant set. Generalizability to other organs or broader clinical use remains untested.
Lin et. al. (2025) [36]	**Auto-scanning for reproducibility (Auto-RUSS):Uses a 7-DOF robotic arm with force control and visual servoing. Evaluated against physicians on thyroid and tumor phantoms.**	**Limited scope:** Only tested for force reproducibility on healthy volunteers. Used a single ultrasound device. No in vivo diagnostic validation yet.
Bamaarouf et. al. (2024) [37]	**Cobot for difficult scans:** Presents a co-robotic assistant for scanning difficult patients (e.g., pregnant women with high BMI). Enhances probe stability and force control.	**Phantom-only testing:** Evaluation limited to simulations. No clinical testing on real patients or broader cases.
Najdovski et. al. (2024) [38]	**Haptics-enabled tele-ultrasound (HaptiScan):** Builds a telerobotic ultrasound system with real-time force feedback. Uses a Phantom haptic interface and UR5 arm. Highlights safety features and remote scanning benefits.	**Lab-based demonstration:** Tested on phantoms in a controlled environment. No human trials or large-scale usability validation.
Du et. al. (2024) [39]	**Survey of robot-assisted ultrasound systems:** A comprehensive review categorizing ultrasound systems by automation level. Covers planning, control, and standardization methods.	**Review only:** No new experiments. Limited by literature at time of writing. Regulatory and clinical adoption barriers not deeply analyzed.
Bi et. al. (2024) [31]	**AI-driven robotic ultrasound (Review):** Annual review discussing AI integration in robotic ultrasound. Highlights control strategies, environment modeling, and autonomy challenges.	**Theory-focused:** Emphasizes future directions and challenges. No new technical or clinical results presented.
Planes et. al. (2025) [40]	**5G tele-robotic scanning trial:** Validates a 5G-based robotic ultrasound system on 64 volunteers. Compares results to conventional scans. High patient satisfaction and clinical comparability.	**Single-operator, ideal setting:** Performance may vary with BMI, patient cooperation, or hardware. Not tested in diverse or critical conditions.
Gifari et. al. (2024) [41]	**Prone echocardiography robot:** Robotic manipulator positioned below patient for cardiac imaging. Evaluated on healthy subjects for basic feasibility.	**Very limited validation:** Tested on one healthy subject. Lower image quality with novice users. Needs broader validation.
Han et. al. (2024) [42]	**Remote exam satisfaction analysis:** Uses survey and structural equation modeling to assess participant feedback after robotic exams. Highlights value perception and communication ease.	**Non-clinical context:** Based on healthy employees. No clinical data or diagnostic outcomes.

TABLE 3. ***(Continued.)*** **Comparison of recent reviews on robotic ultrasound systems (2022–2025).**

Reference	Focus Areas	Limitations
Liang et. al. (2024) [43]	**Long-distance robotic scanning:** Conducts tele-ultrasound over 100 km using 5G. All standard views successfully acquired. Confirms feasibility.	**Small-scale study:** Pilot scale, limited modalities. Increase in scan time and potential workflow challenges when scaled.
Zhang et. al. (2024) [44]	**401-patient remote ultrasound study:** Large cohort evaluation of robotic ultrasound on a rural island. Shows high accuracy, minor diagnostic variance.	**Measurement bias and setup reliance:** Slight discrepancies in kidney size, system dependent on robust network and trained operators.
Tang et. al. (2024) [45]	**Vision-guided autonomous echo:** Uses 3D sensing to detect torso landmarks and servo imaging control for cardiac ultrasound. Autonomous scanning path planning and force control.	**Limited to feasibility:** Demonstrated on phantoms and volunteers. Gel applied manually. Obstacles like ribs not handled automatically.
Greenidge et. al. (2025) [46]	**In vivo GI capsule ultrasound:** Mini rolling robot performs internal 3D ultrasound inside gut using external magnetic guidance. Shows real-time scanning in pigs.	**Preclinical:** Only tested in animal models and phantoms. Human trials pending. Magnetic controller and safety need validation.
Zhang et. al. (2024) [47]	**Autonomous CT-guided liver follow-up:** Aligns prior CT with real-time US using deep learning for follow-up lesion scanning. Demonstrated accurate lesion localization on phantoms.	**Limited validation:** Phantom-only. No patient testing. Limited to known lesion follow-ups, not general diagnosis.
Sun et. al. (2025) [48]	**ARUS for musculoskeletal 3D modeling:** Uses stereo vision, hybrid force control, and deep learning segmentation for MSK recon. Compared results with MRI on phantom.	**Phantom-based only:** Accuracy shown in lab-built models. Needs human testing. Currently constrained to predefined regions.
Jiang et al. (2023) [29]	State-of-the-art robotic ultrasound systems; AI for sonographic control; "language of sonography"	Lacks in-depth analysis of force control and image-driven scanning optimization
Lin et al. (2025) [36]	Development and evaluation of auto-RUSS system for observer consistency	System-specific study; lacks broad review of adaptive control strategies
Huo et al. (2025) [49]	Autonomous eFAST imaging; system integration and experimental trials	Application-specific; lacks coverage of multi-modal sensing and image optimization
Kim et al. (2022) [50]	Soft robotics in ultrasound scanning	Focused on mechanical systems rather than control or imaging strategies
Zhao et al. (2022) [51]	Force control for autonomous ultrasound	Does not deeply cover real-time feedback and image optimization
Ali et al. (2023) [52]	Remote robotic ultrasound systems in telemedicine	Less focus on force adaptation and trajectory planning
Wang et al. (2023) [53]	AI-enhanced imaging quality in robotic scanning	Does not integrate force or haptic control strategies

captured in the original query, such as the work by Roshan et al. [98] on ultrasound automation, which emphasizes autonomous scanning path control using learning-based methods. This highlights the limitations of a title-only search, and we have expanded our methodology accordingly to improve completeness and transparency.

III. WORKFLOW OF AUTONOMOUS IMAGE ACQUISITION

In general practice, region of interest is first located during the scanning step before starting the acquisition process. The most often used data was captured using magnetic resonance imaging and computed tomography. This plan formation is different from the other methods where researchers use the information obtained from the patient in real-time through some sensors. Moreover, in some cases, researchers have also used the acquired ultrasound images for the planning of the probe movement.

Once the path for scanning is finalized, then by using a monitored control over the probe for its orientation and position by a robot, the data acquisition is performed. For the safety of the patient, it is important to keep the probe's contact force under constant observation and control to ensure acoustic coupling. These acquired images are further processed for the 3D reconstruction or the detection of some anatomical structures and have been used for feedback for adjustment of the pose of the probe or for changing the scanning path. Once the acquisition process is complete, this data is used further for many applications, which include anatomical and clinical ones, in addition to being used for the evaluation of the systems aided by the specified metrics (Refer to Figure 2).

IV. AUTONOMOUS ROBOTS FOR ULTRASOUND IMAGE ACQUISITION

A manual handling of the probe is often practiced in conventional 2D-Ultrasound scanning, where the obtained images depict clear anatomical structures that can further be used for biometrics or abnormality detection [83], [84]. The primary challenge for the acquisition of 2D images using autonomous robots is the interpretation of the images obtained and the formation of the strategies to control the robot so that the probe is precisely placed. A robot made up of a manipulator with a parallel link of 6 degrees of freedom and a passive arm with a serial link was invented by Nakadate et al. [78], [79]. It is combined with the algorithms incorporating the real-time processing of images for the automatic acquisition of 2D-ultrasound images of the carotid artery for diagnosis. In the resultant images, anatomical landmarks are detected. Following these landmarks, a scanning methodology is formed for searching the plane of desired images of the carotid artery.

An autonomous screening of the liver was performed by using six degrees of freedom manipulator, which is available commercially [85]. The person's epigastric area is indentified using an RGB image of the person's surface taken by a Web camera and a scanning protocol was created to replicate the clinical 2D-US evaluation of the liver. This study demonstrates the capability of employing robotic systems to autonomously capture useful US images of the liver for assessment. A special cage-like robotic platform was designed to be attached to the bed of the patient to obtain cervix images. This technology can scan down explicitly set routes with three translational degrees of freedom and two rotational degrees of freedom to obtain the cervix's ultrasound images along sagittal and coronal views.

Another cage-like robot with five degrees of freedom was designed for the ultrasonography of the fetus [80]. Here probe can move in the horizontal plane only, whereas height is managed by the passive system. It has two degrees of freedom, and the applied force on the patient is kept constant. Fetus and mother safety is enhanced by introducing safe contact between the patient's surface and probe via passive ultrasound probe design. Moreover, several system constraints were observed, such as the limited scan range and shadow abnormalities in the collected ultrasound images [80]. Furthermore, the cage-like architecture in [80] might cause the patient to become uneasy during the robotic US assessment, which ought to be incorporated into the future mechanical design to increase the clinical acceptability of the system.

A robotic system with seven degrees of freedom for the acquisition of ultrasound images of the breast was designed by Welleweerd et al. [81]. They used a linear probe along with an end-effector where the probe path is pre-determined [81]. Li and Deng [95] designed a framework for developing ultrasound scanning abilities using human performances. They have taken into account the positioning of the probe, ultrasound images, and force of contact in their model design. They used a sample-based strategic plan for making the ultrasound assessment autonomous and made a robust system. Tan et al. [96] designed a completely autonomous and robust robotic device for ultrasound breast scans. This system is comprised of dual robot arms and a probe with a clamping device in addition to a human-computer interface. This system can enhance the diagnostic ability of the ultrasound imaging system while also simplifying the procedure of breast ultrasound scan. Advantages of this system are 1. In the proposed working space there is zero singularity, 2. dual probes in combination with the anti-collision strategy makes the system time efficient, 3. Both arms share the same bracket rails resulting in the space reduction thereby optimizing the size of the system, 4. Increased scanning space range in comparison to the single arm, 5. simplification of the motion control algorithm due to zero effect of translational axes, and as gantry bracket structure is used in the system therefore this enhances the stability, safety and uniformity of the system.

A. CUSTOM DESIGNED TELEOPERATED AUTONOMOUS ROBOTS

There are many specially designed robots. MGI Tech Co. made MGIUS-R3, and AdEchoTech made MELODY for commercial use, and they are teleoperated robots for ultrasound imaging [112]. There are 6 DOFs in the later robotic system, which includes a probe and a force sensor. A probe constructed from plastic is made available for the physician at the remote place to have control of the actual probe. Recently a tele-echographic robot was used for the

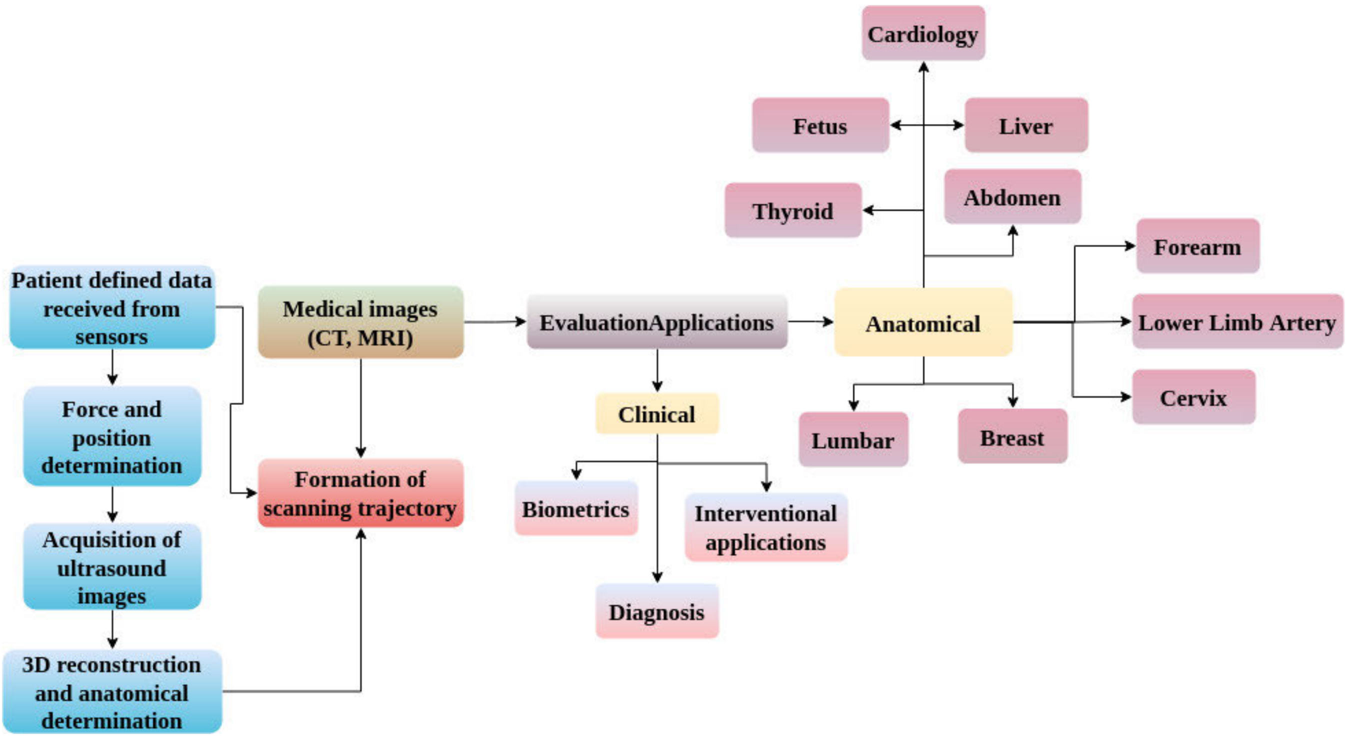

FIGURE 2. Workflow of the autonomous ultrasound robots.

TABLE 4. Summary of 2D ultrasound image acquisition by autonomous robots.

Ref.	Anatomical Applications	Sensors	Manufacturer	Ultrasound device
[78]	Liver	Force sensor with six-axis and Web cam	Mitsubishi Electric, Ltd., Japan (MELFA RV-1) 6 DoF	Hitachi Aloka Medical Ltd., Japan (Diagnostic Ultrasound machine)
[79]	Carotid artery	N/A	Custom Design 6 DoF	ALOKA Co. Ltd. Japan Pro Sound II SSD-6500SV
[80]	Fetus	USL06-H5, Tec Gihan, Japan (3-axis force sensor)	Custom Design 5 DoF (3- Translation 2-Rotation)	General Electric, USA LOGIQ S8 Convex probe CI-5
[81]	Breast	N/A	KUKA LBR Med7 R800 KUKA Roboter GmbH, Germany 7 DoFs	Siemens AF, Germany Siemens X300 Linear probe VF13-5
[82]	Thyroid	RealSense D4Si, Intel Co-operation, USA. RGB-D camera	Franka, Emika, GmbH, Germany (Franka Emika Panda) 7 DoFs	Verasonics, Inc., USA Verasonics Vantage 128, Linear probe ATL L7-4

covid-19 diagnosis [25]. There are six degrees of freedom used in the MELODY system, where three are active and three are passive DoFs. The first three active DoFs are used for probe rotation, and the later three passive DoFs are for probe positioning. In addition, it also has a force sensor. A human assistant operates the robot's coarse translation, whereas the positioning of the probe is controlled precisely using a force feedback haptic device by an experienced sonographer. Practical implementation of MELODY for more than 300 patients includes pelvic, abdominal [113], [114], vascular telesonography [113], cardiac [114], and obstetric [116]. The kinematically robust robotic arm has seven active degrees of freedom (DOF) and an extra force-torque sensor, and it was expressly built to replicate all of the actions of a human examiner [119]. ReMeDi, unlike MELODY, somehow doesn't rely on user intervention. This technology had already been developed and tested in 14 patients for remote cardiac assessments [120]. The TOURS (Tele-Operated Ultrasound System) includes a small robotic probe controller with three active degrees of freedom (DOF), allowing remote control of probe position through a phantom probe without haptic input [121]. An attendant at the patient facility handles translation physically. TOURS has been evaluated across extended distances in over 100 patients for vascular, abdominal, obstetric, and pelvic assessments [121]. The device was also utilized for distant ultrasound scans on the International Space Station with success [122]. A probe phantom was used for probe rotations and an ordinary keyboard for the translational motion to operate a specially developed robot with six degrees of freedom and a force sensor [123]. A strong participant proved the effectiveness. A small parallel telerobotic system with six degrees of freedom for precision probe placement and haptic feedback for mobile was reported in [124].

B. COMMERCIALLY DESIGNED ROBOTS

An ultrasound robot with minimal cost was invented with six degrees of freedom using a universal robot UR5 [88]. An external force sensor was employed to augment the integrated torque data, and a haptic gadget was utilized for remote control. Although the system fits the technical specifications for teleoperated ultrasonography, it has not yet been tested in humans [88]. Research employing the UR5 robot looked into filtering haptic instructions and slowing down to increase safety [125]. A low-cost robot was designed by Mathiassen et al. [88] by using universal robots UR5. The purpose was to reduce the stress of controlling the robot, which is responsible for probe movement. They used Gamma SI-65-5 and Phantom Omni for the force and the haptic control. The system has 6 degrees of freedom. Implementation of a ProSix C4 robot (Epson) lacking force sensors to acquire ultrasound images by controlling the probe remotely using a joystick for the reconstruction of a 3D volume was given in [128]. The camera is used for safety analysis and visual surveillance. Vascular scanning is performed on healthy subjects for testing of their proposed technique. A combination of ultrasound with RGB-D feedback was incorporated in the study by Huang et al., [135]. Integrating these sensor data might allow for more accurate manipulation of the ultrasonic probe and the acquisition of optimal image scans.

A novel control strategy was proposed in [126] and [127] employing a lightweight anthropomorphic robot (WAM, Barrett Technology) with seven degrees of freedom and haptic device remote control. An externally provided force sensor and a camera with 3D time-of-flight were put together to provide a seamless transition process between free movement and patient touch. The design was confirmed in a pelvic exam performed on a healthy volunteer in the exact chamber as the examiner. A support system based on an ultrasound diagnostic robot that can help manual manipulation of ultrasound probes in echography to reduce examiner fatigue was developed by Masuda et al. [129]. High-grade autonomy was achieved by a robotic ultrasound system RobUSt developed by Suligoj et al. [130]. This system can undergo auto-selecting of the anthropomorphic geometry by making use of the data obtained in the three dimensions from the camera. The control over impedance allows the ultrasound probe to be placed while accounting for tissue deformation or surface motion.

V. DISCUSSION

A. CUSTOM MADE OR INDUSTRIAL ROBOTS

The practicality of doing remote ultrasound scans of diverse anatomical locations at variable distances has been demonstrated during the last five years. Patients and examiners usually embrace this new technology, which has the potential to increase access to treatment, for example, by shortening consultation wait times in distant places where expert sonographers are few [131]. Even though the custom-designed robots offered similar capabilities to typical industrial robots and cobots, the usage of the parallel kinematic design is what makes custom designs different from traditional industrial designs. Even though industrial robots and cobots are generally inexpensive, custom-made robots can be confined to executing only the explicitly defined task, potentially lowering expenses only when high numbers are produced. The benefit of deploying a cobot over a standard industrial robot because it already has built-in safety precautions, providing a more effective, safe situation for human interaction. Typical robot systems included safety procedures like visual and force control. Nevertheless, it is unclear if these characteristics make them adequately safe to be used in a healthcare context. To have an autonomous ultrasound system, it's better to use serial kinematic design rather than those commercial designs since it has a smaller ratio of the workforce to footprint ratio. We should highlight the opportunity which comes with the new affordable robot-arms which are originally built for industry applications. These devices provide features like force and torque control which makes them suitable for application in ultrasound diagnostics. Force-sensing devices use either the robot's inherent sensor or external ones. Since 2015, the use of 3D cameras has become increasingly popular. The visual depth perception obtained with 3D cameras enables the system to construct the probe's route for deception in 3D space and is also used in vision-based controls. The main features of Microsoft Kinect, such as its cost-efficient, real-time depth sensing, and RGB image acquisition, make it the most popular option amongst 3D cameras [133]. In addition, some studies also used INTEL RealSense 3D cameras. These are more costly than the Microsoft Kinect; however, it provides more resolution, field view, depth range, and frames per second [134].

At the time of optimization and scanning, visual control is handled by using RGB-D, ultrasound and feedback. Probe manipulation is performed using ultrasound visual control in earlier studies. Later on, the investigations moved their emphasis to the use of RGB-D sensors. The RGB-D sensor may give additional external spatial information about the positioning of the probe. Moreover, ultrasound image processing can determine interior organ anatomy with quality images. A combination of ultrasound with RGB-D feedback was incorporated in the study by Huang et al., [135]. Integrating these sensor data might allow for more accurate manipulation of the ultrasonic probe and the acquisition of optimal image scans. Few studies made use of RGB-D visual control. The disadvantage of this method would be that the probe may only be moved to a predetermined depth below the surface of the region of interest. As a result, irrespective of a patient's gender, shape, or age, the scanning depth stays constant, potentially affecting the quality of the obtained images as well as the patient's well-being.

TABLE 5. Summary of 3D ultrasound image acquisition by autonomous robots.

Ref.	Clinical Application	Anatomical Application	Sensors	Manufacturer & Robot Model	Ultrasound device
[86]	Biometric	Lower limb arteries	Industrial Automation, USA (6-axis Force/torque sensor)	CRS Robotics Co., Canada F3 articulated robot 6 DoFs	Analogic Co., USA SonixTouch Linear probe L14-7
[87]	Biometric	Abdominal aortic aneurysms	Kinect, MS Co., USA. RGB-D camera	KUKA Roboter GmbH, Germany. KUKA LBR iiwa R800 7DoFs	Analogic Co., USA. Sonix RP Convex Probe 4DC7-3/40
[88]	General	General	Gamma SI-65-5 ATI Industrial automation 6 DOF Phantom Omni (3 Active DoF, 3 Passive DoF)	Universal Robots UR5	VGA2 Ethernet Epiphan
[89]	Biometric	Thyroid	Intel Co., USA. RealSense F200 RGB-D Camera	KUKA AG, Germany, KUKA LBR iiwa7 R800, 7DoFs	Analogic Co., USA, SonixTable Linear Probe L14-5
[90]	Interventional	General	Kinect, Microsoft Co., USA, RGB-D camera	KUKA Roboter GmbH, Germany, KUKA iiwa 7DoFs	Analogic Co., USA Support Sonix RP Convex Probe 4DC7-3/40
[91]	Interventional	General	Intel Co., USA, RealSense SR300 RGB-D Camera	KUKA Roboter GmbH, Germany, KUKA LBR iiwa7 R800 7DoFs	Analogic Co., USA SonixTable Convex Probe C5-2/60
[92]	General Purpose	General	Kinect, Microsoft Co., USA RGB-D camera	KUKA Roboter GmbH, Germany. KUKA iiwa LBR5 7DoFs	Analogic Co., USA Ultrasonix Convex Probe C5-2
[33]	General Purpose	General	Kinect, Microsoft Co., USA RGB-D camera, IMS-Y-Z03, I-Motion Inc., China	Leetro Automation Ltd., China 3D translating device	Analogic Co., USA Sonix RP Linear Probe L9-4/38 and a convex Probe C7-3/50

TABLE 6. Projects of robot-assisted echography.

Ref.	Year	Mode	Country	Title
[97]	1998	Teleoperated	European Union	MIDSTEP
[99]	1998	Teleoperated	France	Hippocrate: Medical Image Analysis
[60]	1999	Teleoperated	France	Hippocrate Medical Image Analysis
[59]	2001	Teleoperated/ Autonomous	United States	TER/TERMI
[100]	2001	Teleoperated	Japan	RUDS
[101]	2002	Teleoperated/ Autonomous	European Union	In vivo
[102]	2003	Teleoperated	European space agency	TERESA: Space research to ground tele-echography
[103]	2004	Teleoperated	European Union	OTELO
[105]	2007	Teleoperated	European space agency	TERESA Abdominal echography
[106]	2007	Teleoperated/ Autonomous	United States	TER/TERMI
[105]	2007	Teleoperated	European space agency	ESTELE
[107]	2010	Teleoperated Human-Robot Cooperation	France	PROSIT
[108]	2010	Autonomous	Japan	WTA-2
[109]	2011	Teleoperated	German	FP-USM
[110]	2019	Autonoumous	United Kingdom	iFIND
[111]	2019	Autonoumous	United Kingdom	iFIND

Occurring point cloud noises retrieved from RGB-D, may prevent a good connection between the surface of the patient and probe [136]. The variable elastic characteristics and tissue stiffness could be used to manipulate the force-based ultrasonic probe. Furthermore, even in the absence of visual input, the force components may be used to autonomously position the probe normal to the surface using orientation optimization approach [137]. The use of these technologies for scanning and optimization might overcome visual control restrictions such as reliance on ambient lighting and patient involuntary motions. In manual ultrasonography, an ultrasound probe can be moved in six degrees of freedom (DOFs) during scanning and image optimization to execute fundamental ultrasound probe motions. Some studies used six degrees of freedom for the positioning of the probe [78], [79], [86], [88], [92], [135], [136], [138], [139], [140], [141], [142], [143], [144]. The studies where the degrees of freedom are less than six undergo more limitations for probe movement. As a result, while applying several ways to provide a good image, these systems may not provide an excellent view of the intended structure. This may result in a misunderstanding of the structure (fragments) or an incorrect diagnosis.

Recent advancements in robotic ultrasound scanning have explored methods to enhance probe positioning accuracy and image quality. One significant challenge in robotic ultrasound imaging is compensating for patient motion and ensuring precise probe repositioning. Jiang et al. [145] proposed an approach for precise repositioning of robotic ultrasound probes by improving registration-based motion compensation using ultrasound confidence optimization. This method refines the positioning process, ensuring better stability and consistency in imaging.

Another critical aspect of robotic ultrasound is automatic normal positioning of the probe, which traditionally requires manual adjustments or external tracking systems. Jiang et al. [146] introduced an automatic positioning method based on confidence map optimization and force measurement, eliminating the need for additional external tracking devices. Their approach enhances the adaptability of robotic ultrasound systems by using only intrinsic data from the ultrasound probe, making it more efficient and robust for real-time applications.

In all of the suggested solutions, safety procedures in the case of sensor or robot malfunction were not considered. These risks can be mitigated by implementing redundancy in systems that provide critical functionality, such as contact force estimates. While redundancy may be provided by introducing redundant sensors, enhanced functionality could also be implemented to allow current sensors to be utilized for additional objectives. For instance, contact force might be measured by utilizing a force sensor between the robot and the ultrasonic probe, as well as a joint measure of torque or motor current combined with kinematic computations. Likewise, calculating robot velocity with accessible camera systems might be used to validate the velocity computed by the robot sensors. Moreover, to sensor redundancy, all active features must be built to necessitate power, whereas the secure state ought to be that without power. Furthermore, computer system redundancy is another critical component. Other mechanical safety measures include hardware brakes that restrict the robot's attainable workspace and the use of clutches in joints that are released during emergency braking.

VI. FORCE CONTROL MECHANISMS IN ROBOTIC ULTRASOUND

Precise force control is fundamental in robotic ultrasound systems to ensure patient safety and consistent image quality. Excessive contact pressure can cause patient discomfort or injury, while too little pressure degrades image clarity [54]. Modern robotic probes therefore integrate force-torque sensors to monitor contact forces in real time. Many recent systems mount a six-axis force/torque sensor at the probe, which measures three orthogonal forces and three moments at the probe-tissue interface [35]. This provides not only the normal (axial) contact force but also lateral (surface-parallel) force components and tilting torques. Such multi-axis sensing allows the controller to detect any shear forces or uneven contact and adjust accordingly. For instance, maintaining the probe normal to the skin surface (minimizing off-axis torques) is known to improve ultrasound image quality by ensuring uniform acoustic coupling [55]. To achieve this, controllers use torque feedback to actively correct probe orientation, reducing lateral shear and keeping the transducer perpendicular to the body. Recent designs have even integrated custom force sensors into the probe attachment to reduce size and cost, enabling widespread use of force feedback. Zheng et al. [54] demonstrated a lightweight 3D-printed force sensor embedded in the probe holder, combined with a hybrid position–force controller to maintain optimal probe contact. This sensor provided reliable force measurements for closed-loop control while avoiding the expense and bulk of a traditional six-DOF transducer.

Keeping a consistent normal force during scanning is critical for image consistency. Studies have shown that constant contact force is vital for high-quality imaging, as fluctuations in pressure can alter tissue deformation and image properties [36]. Accordingly, robotic systems employ force controllers (often based on impedance or admittance control) to regulate probe pressure. A target force in the range of a few newtons is typically maintained – for example, Su et al. kept contact force between 2–4 N in an autonomous thyroid scan to ensure sufficient contact without distorting the tissue [35]. Advanced force control strategies can adapt to the compliance of human tissue. Traditional fixed-gain controllers may overshoot, causing unsafe spikes when the probe first contacts the body [56]. To address this, recent works introduce adaptive compliance control. Jiang et al. proposed an integral adaptive admittance control that estimates soft-tissue stiffness online and modulates the robot's motion to avoid force overshoot and maintain a stable force during scanning [56]. By dynamically adjusting controller parameters, this method kept contact pressure steady on uncertain, deformable surfaces, improving patient safety and image stability. Other groups leverage learning-based approaches to refine force control. Ning et al. developed a reinforcement learning agent that uses only force sensor input to control the probe's contact posture and force, achieving stable contact on a soft phantom without prior knowledge of the environment [57]. The learned policy, combined with an admittance controller, enabled synchronous force and orientation control to keep the probe pressed at the desired force while adapting its angle to the surface [57]. This resulted in more reliable scanning on uneven anatomy, as the probe could actively adjust to maintain both pressure and stability. Overall, by integrating multi-axis force/torque sensing with advanced control (adaptive feedback loops, compliant mechanisms, or even AI controllers), robotic ultrasound systems can ensure a constant, appropriate contact force. This improves safety and image consistency, as evidenced by higher image quality and reduced variability when force is well-regulated [35], [36]. Modern force-controlled probes thus protect the patient from excessive pressure while guaranteeing the acoustic coupling needed for reproducible, high-quality ultrasound images.

A. EXISTING METHODS FOR ADDRESSING CHALLENGES IN ROBOTIC ULTRASOUND

Various control and optimization techniques have been employed in robotic ultrasound systems to improve precision, adaptability, and real-time response. Among the widely utilized methods are:

1) ASYMPTOTIC TRACKING WITH INTEGRAL ROBUST CONTROL FOR UNCERTAIN NONLINEAR SYSTEMS

Many robotic ultrasound systems encounter mismatched uncertainties due to variations in soft tissue properties, patient movement, and unpredictable probe-skin interactions. Asymptotic tracking control, particularly with integral robust schemes, has been successfully applied to mitigate these uncertainties. These methods ensure that the robotic system can maintain precise trajectory tracking while compensating for disturbances and unknown dynamics in real time.

2) MODEL PREDICTIVE CONTROL (MPC) FOR FORCE AND POSITION OPTIMIZATION

MPC has been extensively used in robotic-assisted ultrasound to optimize probe force and positioning dynamically. By predicting future states based on current and past measurements, MPC enhances probe stability and ensures optimal contact force, which is crucial for high-quality imaging.

3) ADAPTIVE AND LEARNING-BASED CONTROL APPROACHES

Recent advancements in machine learning and adaptive control have introduced AI-driven methods that adjust scanning parameters based on real-time feedback. Reinforcement learning, for example, has been applied to optimize probe positioning by learning from previous interactions with different tissue surfaces.

4) HYBRID FORCE-POSITION CONTROL FOR IMPROVED PROBE MANIPULATION

Many studies incorporate hybrid control strategies that combine force and position feedback to achieve stable and consistent probe contact with the skin. These methods often utilize force-torque sensors and RGB-D data to enhance image quality and reduce errors in probe orientation.

5) ULTRASOUND IMAGE-GUIDED OPTIMIZATION ALGORITHMS

To ensure high-quality imaging, some studies integrate real-time ultrasound image feedback into the optimization process. Techniques such as edge detection, texture analysis, and deep learning-based segmentation are utilized to refine probe positioning and trajectory adjustments dynamically.

B. SAFETY: POSITION CONTROL (VISUAL/CAMERA) AND FORCE

At present patient cannot be left alone with a robot performed the ultrasound scan. Medical staff should always be available to assist the patients and - if necessary to remove the probe from the chest. Control of the probe position is possible by a sonographer standing next to the patient using remote probe movements via a 3D mouse or joystick attached to the ultrasound scanner. In this paragraph you need to include visual control of the probe position by a sonographer next to the patient. That is what we are currently doing. For tete-operation control of the probe position can be performed by sonographers using a camera. However, there must be at least one other medical staff close to the patient.

C. ALGORITHMS FOR ADJUSTING THE POSITION OF THE PROBE NEEDS TO BE INCLUDED

This is probably the most challenging task. As suitable and affordable hardware (robot arms, cameras, ultrasound scanners) are available, software development to connect the components becomes the major objective to advance autonomous ultrasound imaging. Like an experienced sonograpger, an autonomous robot has to adjust its position according to patients movement and breathing as well to the analysis of the ultrasound imaging which are recorded continuously. Decisions have to be made, whether the images are adequate. If not, modifications of the position of the arm have to executed and all these actions should happen with minimal time delay. The critical part is the automated assessment of ultrasound images. That is easier for abdominal organs which remain in a stable position during breathhold, but can be very changing for ultrasound exams of the heart.

VII. QUALITY AND LIMITATIONS

The quality evaluation used in this review was built on thorough and established quality assessment criteria. This guaranteed that the included research was of high quality and that the reader received a comprehensive evaluation of the present status of autonomous robotic systems. Numerous studies may have been ignored since they were included depending on the search technique. For example, research that suggested automation systems but did not include robot-related terminology in the title or abstract may have been disregarded. There are numerous obstacles to using automated systems for various healthcare services and initiatives. This critic probably also relates to the studies included in this review. The limitations related to data include its systemic bias, due to low quality assessment of self reporting there are questions about the reliability, accuracy and validity of the data. The data acquired would be in most of the cases unstructured and raw, which causes difficulty in its analysis. Another issue is the feasibility of the data acquisition and cleansing. There is a risk of robot malfunction, nerve damage and compression.

VIII. CONCLUSION

The review provides thorough information on several technologies utilized in autonomous robotic systems for medical imaging applications. Cobot-based systems are currently receiving greater attention because of their built-in sensor features and compliance with human cooperation criteria. A significant number of devices that utilized an RGB-D sensor were able to execute positioning and preliminary approach without the assistance of a sonographer. Those systems were primarily responsible for the initial strategy and scanning monitoring via path creation. Employing real-time feedback to update the scan route would enhance accuracy and adjust for participant motions. A substantial number of systems employ scanning and optimization with force and RGB-D feedback. Force feedback was primarily utilized to maintain probe interaction with the region of interest while delivering a preset force. Most of the studies made use of the force-torque sensors with a 6-axis. However, they did not consider the surface parallel force components as well as the respective torque components during the control or scanning procedures. Incorporating these force components in the studies might be able to enhance the probe positioning and orientation in addition to the manipulation. Despite the use of numerous optimization approaches, adding ultrasonic image feedback into the optimization process is the most effective way to guarantee that the image acquired is exact and of high quality. Including the study of widely available ultrasound images may thereby increase the accuracy and resilience of these systems.

Most prior surveys of robotic ultrasound have offered broad technological overviews without delving into the nuances of force control or closed-loop feedback, a limitation noted in their own brief picture of the field [29]. In contrast, very recent studies (2022–2025) have begun to tackle these specifics in depth. Modern robotic ultrasound platforms now commonly integrate multi-axis force/torque sensing to regulate probe–patient contact with high precision [35]. Building on this hardware, researchers have developed advanced adaptive control strategies: for example, an integral adaptive admittance controller was shown to avoid force overshoot and maintain stable contact pressure on soft tissue [56], and a learned compliance control scheme (via inverse reinforcement learning) enables simultaneous adjustment of probe posture and contact force in unstructured scanning conditions [36]. Such controllers operate in real time and are often paired with feedback-based path correction mechanisms. For instance, one recent system dynamically adjusts the probe's scanning trajectory in response to patient motion and force feedback, ensuring consistent imaging despite disturbances [56]. Equally significant is the integration of visual feedback (e.g. ultrasound image guidance) into the control loop. Vision-based methods now incorporate ultrasound image analysis on-the-fly to guide probe positioning e.g., using real-time organ segmentation to correct the probe orientation or location, which dramatically improved measurement accuracy in a thyroid ultrasound task [36]. Another approach decouples control objectives by using ultrasound image feedback to steer the probe's viewing angle while force feedback concurrently governs probe pressure, achieving robust autonomous vascular imaging with minimal human intervention [36]. Collectively, these recent works provide the detailed analyses of multi-axis force control and real-time feedback integration that earlier broad reviews lacked, directly addressing previous gaps by demonstrating how closed-loop force regulation and adaptive, feedback-driven path planning can enhance imaging consistency and safety [29].

A. CONTRIBUTION OF SURFACE-PARALLEL FORCE COMPONENTS AND TORQUE TO PROBE CONTROL

A key focus of this review is the analysis of how surface-parallel force components and torque influence the mechanical behavior of robotic ultrasound probes during scanning. Traditional force control in robotic ultrasound systems has centered predominantly on regulating the normal (axial) force applied by the probe to the patient's skin, ensuring sufficient coupling for image quality without exceeding safety thresholds [36], [39]. However, this approach neglects lateral force vectors and the rotational moments (torque) that are critical for maintaining consistent orientation and minimizing probe slippage—particularly on curved or compliant anatomical regions.

Recent studies have highlighted that uncontrolled surface-parallel forces can lead to asymmetric tissue deformation and motion artifacts in the captured ultrasound images [35]. For instance, Bamaarouf et al. [37]noted that maintaining lateral stability was essential in co-robotic systems designed for obstetric imaging, where non-uniform body surfaces and probe orientation heavily impact image clarity. Similarly, Najdovski et al. [38] integrated 6-axis force/torque sensing in their HaptiScan platform to monitor and mitigate both translational shear and unintended rotational drift of the probe during teleoperated scans.

Surface-parallel forces—those acting tangentially to the scanning plane—require real-time compensation to avoid skewed or tilted probe positions, which can otherwise degrade acoustic coupling and introduce shadowing or dropout artifacts. Several modern systems have implemented torque-aware controllers that leverage lateral force data to actively stabilize the probe's roll and pitch angles relative to the skin surface [45], [46]. This rotational control is especially beneficial in tasks like vascular scanning, where even minor angular deviations can displace the acoustic window from the target.

Moreover, learning-based controllers have begun to factor in torque signatures to guide dynamic compliance. For example, Sun et al. [48] incorporated both shear force and torque feedback into a policy trained via reinforcement learning to stabilize probe contact across variable patient geometries. Their results demonstrate that accounting for these additional mechanical components enables smoother force profiles and improved diagnostic image stability during scanning.

Thus, our review provides a synthesized investigation into these underexplored control dimensions. We identify that only a limited number of studies explicitly decouple and analyze surface-parallel forces and torque in robotic probe manipulation. Incorporating these into control strategies—alongside axial force—can enhance probe orientation robustness, prevent probe rotation or lateral drift, and ensure consistent pressure application throughout the scan. Addressing these aspects is essential for achieving reliable and repeatable autonomous and semi-autonomous robotic ultrasound performance.

B. ULTRASOUND IMAGE-BASED OPTIMIZATION FOR IMPROVED DIAGNOSTIC PRECISION

In robotic ultrasound imaging, accurate probe placement alone is not sufficient to ensure diagnostic-quality images. Instead, continuous optimization based on the acquired ultrasound image content is essential to achieve precise anatomical localization and optimal acoustic windowing. This approach, referred to as ultrasound image-based optimization, involves real-time adjustment of the probe's position, orientation, or contact force based on metrics extracted from the ultrasound images themselves. The objective is to maximize image quality and anatomical relevance, thereby improving diagnostic precision.

Recent systems have demonstrated various techniques to integrate ultrasound image feedback directly into the control loop. A common method involves using quantitative image quality metrics such as entropy, brightness, sharpness, or confidence maps as optimization objectives. For example, Su et al. [35] employed Bayesian optimization to iteratively adjust the probe orientation and maximize the entropy of thyroid ultrasound images. This allowed the robot to autonomously identify the angle that produced the clearest and most information-rich view of the thyroid gland. Similarly, Zhang et al. [47] utilized a real-time segmentation model to extract the region of interest (ROI) from the ultrasound video stream and adjusted the probe trajectory to keep the ROI centered and maximally visible.

Another emerging technique is the use of deep learning models to predict diagnostic relevance from the ultrasound stream. Greenidge et al. [46] trained a convolutional neural network (CNN) to score frames based on clinical usefulness. This score was used as the reward signal in a Bayesian controller that selected new scanning trajectories. The result was a robot that could explore a scanning area, learn which poses yielded clinically usable images, and automatically adjust the probe to improve overall image quality and coverage.

More advanced systems have employed reinforcement learning (RL) to learn closed-loop scanning policies. In these systems, the RL agent receives feedback from both force sensors and ultrasound image analysis networks. The reward function often incorporates anatomical detection accuracy or segmentation overlap metrics, which encourages the agent to adjust its motion in ways that improve visibility of target structures. For example, the ARUS framework by Sun et al. [48] combines stereo vision, force feedback, and real-time ultrasound segmentation to reconstruct 3D models of musculoskeletal anatomy. During scanning, the robot adjusts its path to improve segmentation quality frame-by-frame, demonstrating how image-driven feedback can guide dynamic optimization of both probe trajectory and contact force.

Incorporating ultrasound image feedback into the control architecture allows the robot to behave more like a trained sonographer: adjusting its probe manipulation strategy based on what it *sees* in the image. This feedback loop enables real-time correction of off-angle views, poor coupling, or probe drift, and ensures that the image being acquired meets diagnostic standards. Moreover, this reduces inter-operator variability and increases reproducibility, which are essential for standardizing ultrasound imaging in clinical workflows.

In summary, this paper highlights the transition from pre-defined scanning paths to intelligent, adaptive strategies where the ultrasound image actively informs probe control. These image-based optimization methods, particularly those using machine learning, not only improve image quality but also enable consistent capture of diagnostically relevant views—an advancement that is pivotal for both autonomous and operator-assisted robotic ultrasound systems.

IX. FUTURE RESEARCH DIRECTIONS

Despite recent progress in robotic ultrasound technologies, several open challenges and unexplored opportunities remain. Addressing these gaps will be crucial for advancing the clinical translation, autonomy, and diagnostic robustness of robotic ultrasound systems. Based on our synthesis of current literature, we identify the following key directions for future research:

A. ENHANCED IMAGE-GUIDED CONTROL ALGORITHMS

Although several systems incorporate real-time image feedback, future work should focus on integrating advanced deep learning models—such as attention-based CNNs, transformer architectures, and uncertainty-aware networks—to interpret ultrasound images more accurately during scanning. These models can support both real-time segmentation and quality assessment, enabling the robot to adapt probe orientation, depth, and trajectory dynamically. Moreover, combining image-based metrics (e.g., entropy, focus, anatomical confidence) with control signals in closed-loop architectures can significantly improve diagnostic view consistency across patient populations.

B. REAL-TIME ADAPTIVE CONTROL STRATEGIES

Robust real-time control remains a bottleneck, especially in dynamic or unpredictable scanning conditions. Reinforcement learning, inverse reinforcement learning, and imitation learning offer promising frameworks for learning adaptive probe control policies that can generalize across different tissue types, patient movements, and anatomical targets. Future research should also explore hybrid control approaches that integrate model-based controllers with learned policies, allowing for both explainability and flexibility in real-world applications. Multi-task learning across scanning protocols (e.g., abdominal, cardiac, thyroid) may also improve generalization and reduce training time.

C. HAPTIC FEEDBACK AND TELEOPERATION ENHANCEMENTS

Current teleoperated systems often lack rich tactile feedback, limiting the operator's ability to assess probe–tissue interaction remotely. Future systems should incorporate high-fidelity haptic rendering of contact forces, including normal and surface-parallel components, to provide more intuitive control for remote sonographers. Additionally, research should focus on minimizing latency and ensuring stable force transmission over wireless or 5G networks. Coupling haptics with semi-autonomous features such as auto-alignment or force stabilization can enhance safety and precision in tele-ultrasound.

D. CLINICAL VALIDATION AND STANDARDIZATION

To move beyond laboratory demonstrations, robotic ultrasound systems require rigorous clinical validation. Large-scale, multi-center trials across diverse demographics and clinical indications are essential to benchmark diagnostic performance, operator usability, and patient outcomes. Furthermore, standardization efforts are needed to define safety thresholds (e.g., contact force, scan time), reporting guidelines, and interoperability protocols with existing hospital infrastructure. Collaborative efforts among researchers, clinicians, and regulatory bodies will be critical to establish medical-grade certifications and clinical workflows.

E. MULTI-MODAL SENSING AND FUSION

Combining ultrasound with complementary sensing modalities can enhance diagnostic context and improve scan robustness. Examples include integrating elastography to assess tissue stiffness, using RGB-D sensors for body surface reconstruction and landmark detection, or fusing inertial data to compensate for patient motion. Moreover, predictive modeling (e.g., AI-based tissue deformation estimation) could support proactive trajectory planning in real time. Future research should explore standardized fusion frameworks that leverage both spatial and temporal alignment of these heterogeneous data streams.

F. USER-CENTERED DESIGN AND HUMAN-ROBOT COLLABORATION

Involving end-users—radiologists, sonographers, and patients—in the design loop is critical to improve acceptance and usability. Future systems should offer customizable levels of autonomy and interfaces that support natural interaction, such as voice or gesture-based control. Additionally, studies on cognitive workload, learning curves, and task delegation in shared autonomy scenarios will help refine the roles of human operators and robotic intelligence in clinical workflows.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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